

ALG-CLTAV-01 wrapper

CL-TAV specification (CR-2026-013 / DD-033)

This wrapper exists only to compile `ALG-CLTAV-01.tex`. It is not a second algorithm. The procedure is the typeset form of the method-report contract. It is not an executable engine and does not select solver technology. Step labels S0–S10 are the stable bilingual mapping; typeset line numbers are not cited.

Chinese step groups (same process, not a second program)

- S0: Γ 为唯一状态源; q / `retry` / $R_{\max}$ 是 $\Gamma.qStatus$ / `retryCount` / `retryCap` 的别名; 初始化 η 、 H_0 与 `retry`。
- S1: 停止 / 错误优先门; 已决定的 A2–A5 或 P1–P5 立即返回。
- S2: 选择前预测; `PredictionGapError` 只关掉 TEST 评分; UNKNOWN 且 Recover 仍合格时不得挡合法恢复。
- S3: 互斥选择 TEST minimax / `Prep` / `Recover` / `Admit`; 空 TEST 集不得 `arg min`; 空 $\mathcal{C}$ 不阻止合格 `Recover`。
- S4: 一次计费; 重试上限不是预算耗尽。
- S5: 发出已准入动作并记录。
- S6–S7: 执行 ERROR、解释 ERROR、`prepErr` 与未确认 `Recover` 汇入同一计数通路; 不排除 H ; η 标为保守未知; 达 $R_{\max}$ 返回 `Stop-Error`。
- S8: 仅 `confirmed = true` 把 $\Gamma.qStatus$ 写成 KNOWN (声明目标与依据); 未确认不经 S10。
- S9: 有效观测或已确认 `Recover` 才更新 η 与 H ; 下一轮 S2 只读 Γ 与 η 。
- S10: 互斥 P1–P5; 未确认恢复回边在 S7, 不在此停止家族。

Stop-645 在本增量之后表示仍未在能力/实现层关闭的具名 645 依赖义务, 不是“645 文件缺失”。

Algorithm 1: CL-TAV top-level session (ALG-CLTAV-01)

Require: declared $H_0 = \{h_{\text{normal}}\} \cup H_{\text{single}}$; SessionContext Γ (unique state); HistoryHandle η ; resource mode $\text{BUDGET}(c_{\min}, B) \text{ xor } \text{ROUNDS}(K_{\max})$; action library; measurement uncertainty

Ensure: stop class, final H , η , session trace; singleton h_{normal} is not protocol PASS

// S0 $H \leftarrow H_0$; aliases $q := \Gamma.q\text{Status}$, $\text{retry} := \Gamma.\text{retryCount}$, $R_{\max} := \Gamma.\text{retryCap}$; $\Gamma.\text{retryCount} \leftarrow 0$; q UNKNOWN unless declared; η empty handles

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1 while true do
  // S1 stop/error gate before prediction
2  if IF-RESOURCE-STOP( $\Gamma, H, \eta$ ) already decided Admit A2-A5 or P1-P5 then return that stop,  $H$ ,  $\eta$ , trace;
  // S2 predict before select; PredictionGapError is TEST-scoped
3  ( $\text{pred}, C$ )  $\leftarrow$  IF-PRED-OBS( $\Gamma, \eta, H$ , uncertainty, library);
4  if  $\text{pred}$  is PredictionGapError then
5    TEST scoring unavailable;  $C \leftarrow \emptyset$ ;
6    if Recover/Prep not eligible from  $\Gamma$  then return named prediction-gap; do not score empty  $C$  as remaining  $|H|$ ;
7   $A \leftarrow$  currently enabled actions;
  // S3 exclusive kind; empty  $C$  does not block eligible Recover
8  ( $\text{kind}, t^*, S, \text{admit}$ )  $\leftarrow$  IF-SELECT-ADMIT( $A, \Gamma.q\text{Status}, C, \Gamma, \eta, H$ );
9  if  $\text{kind} = \text{ADMIT}$  then
10   return  $\text{admit} \in \{A2, A3, A4, A5\}$ ,  $H$ ,  $\eta$ , trace;
11  else if  $\text{kind} = \text{TEST}$  then
12   if  $S = \emptyset$  then return named spec error; do not arg min empty  $S$ ;
13    $t^* \leftarrow$  one-step minimax on currently valid nonempty classes  $C$  inside TEST-only  $S$ ;
14  else if  $\text{kind} \in \{\text{PREP}, \text{RECOVER}\}$  then
15   keep the Prep or Recover  $t^*$  named by IF-SELECT-ADMIT; do not enter TEST minimax;
16  else
17   return named spec error;
  // S4 charge once; retry cap  $\neq$  budget exhaustion
18  charge  $t^*$  once under the selected mode; refuse unadmitted issue;
  // S5 execute
19  ( $\text{record}, \text{execErr}$ )  $\leftarrow$  IF-EXECUTE-RECORD( $\Gamma, t^*$ );
  // S6 interpret, or skip to unified ERROR
20   $\text{obsErr} \leftarrow \text{nil}$ ;  $I_z \leftarrow \text{nil}$ ;
21  if  $\text{execErr}$  is ERROR then
22   fail  $\leftarrow$  true;
23  else
24   ( $I_z, \text{obsErr}$ )  $\leftarrow$  IF-OBS-INTERPRET( $\Gamma, \text{record}$ );
25   fail  $\leftarrow$  ( $\text{obsErr}$  is ERROR);
26   $\text{confirmed} \leftarrow \text{false}$ ;  $\text{prepErr} \leftarrow \text{nil}$ ;
27  if  $\text{kind} \in \{\text{PREP}, \text{RECOVER}\}$  and not fail then
28   ( $\text{confirmed}, \text{prepErr}, \text{declaredTarget}, \text{evidence}$ )  $\leftarrow$  IF-PREP-RECOVER( $\Gamma, \eta, t^*, \text{record}$ );
29  if fail or  $\text{prepErr}$  or ( $t^*$  is Recover and  $\text{confirmed}$  is false) then
30   // S7 unified ERROR: exec, interpret,  $\text{prepErr}$ , unconfirmed Recover
31   do not exclude  $H$ ; mark  $\eta$  conservative-unknown (old KNOWN is not confirmed present);
32    $\Gamma.q\text{Status} \leftarrow \text{UNKNOWN}$ ;
33    $\Gamma.\text{retryCount} \leftarrow \Gamma.\text{retryCount} + 1$ ;
34   if  $\Gamma.\text{retryCount} \geq \Gamma.\text{retryCap}$  then return Stop-Error; this is not A5/Stop-Budget;
35   continue after the one charge;
36  else if  $t^*$  is Recover then
37   // S8 confirmed writes KNOWN( $\text{declaredTarget}, \text{evidence}$ ) only
38    $\Gamma.q\text{Status} \leftarrow \text{KNOWN}(\text{declaredTarget}, \text{evidence})$ ;
39  if  $I_z$  is a valid observation class or Recover is confirmed then
40   // S9 history update; next IF-PRED-OBSreads  $\Gamma$  and  $\eta$ 
41   ( $\eta, H$ )  $\leftarrow$  IF-HIST-UPDATE( $\eta, H, t^*, \Gamma.q\text{Status}, C, I_z$ );
  // S10 exclusive P-stops; unconfirmed Recover does not arrive here
42  let  $\sigma \leftarrow$  first of Stop-Empty, Stop-Singleton, Stop-645, Stop-Equivalent, Stop-Budget via IF-RESOURCE-STOP( $\Gamma, H, \eta$ ) and IF-EQUIV( $\Gamma, H, \eta$ );
43  if  $\sigma$  is set then return  $\sigma, H, \eta$ , and the trace;

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